MEGHARAJ V S

Data Scientist | AI Engineer

Trivandrum, Kerala | +91 7025654877 | megha042023@gmail.com | github.com/megharaj1997 | linkedin.com/in/meghavijayabalan | portfolio-megha-vijayabalan.vercel.app

PROFESSIONAL SUMMARY

Data Scientist with 1.3+ years of production experience building LLM-powered AI agents, multi-path RAG pipelines, and cloud-native ETL systems across a network of 6,000+ EV chargers, serving 1,700+ daily active users and processing 3.5M+ transactions. Proven track record delivering anti-hallucination safeguards, role-based access governance, real-time streaming, and ML-driven pricing in safety-critical, high-throughput environments. Skilled in Python, LangChain, LangGraph, OpenAI API, AWS, and production-grade MLOps, with a strong record of designing and shipping reliable, scalable AI systems that drive measurable business impact.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript

Databases: PostgreSQL, MongoDB, MySQL, Redis

Machine Learning: Scikit-learn, XGBoost, Random Forest, Facebook Prophet, TensorFlow, PyTorch, Keras, NLP, Computer Vision, Time-Series Forecasting, Clustering (KMeans), Feature Engineering

LLMs & AI Systems: LLM Tool Calling, RAG (BM25 + ChromaDB + Cross-Encoder Reranking), Prompt Engineering, LangChain, LangGraph, OpenAI API, AWS Bedrock (Claude), LLaMA/PEFT, Generative AI, Agentic Systems

Analytics & Visualization: Power BI, Matplotlib, Seaborn, Statistical Modeling, Time Series Analysis, Forecasting, PySpark, Parquet

Cloud & Data Engineering: AWS (S3, EC2, Glue, Athena, Bedrock), ETL Pipelines, Apache Spark, Serverless Architecture, FastAPI, Flask, REST APIs

MLOps & DevOps: Docker, Git, CI/CD Deployment Gates, Model Versioning, Redis Caching, Circuit Breakers, Rate Limiting, Regression Testing, Observability & Tracing

Leadership & Methods: Agile/Scrum, Sprint Planning, Cross-functional Team Leadership, Technical Documentation

PROFESSIONAL EXPERIENCE

Data Scientist — ChargeMOD, Trivandrum, Kerala **May 2025 – Present**

Customer Support AI Agent

- Architected a modular monolith support platform with a three-path LLM reasoning pipeline (direct log-based diagnosis, hybrid BM25 + ChromaDB RAG with cross-encoder reranking, dynamic guided questioning), reducing average issue resolution time by approximately 50%.
- Built an entity-cache validation layer that verifies charger IDs and wallet balances against live API responses before LLM injection, structurally preventing hallucination in a safety-critical environment where the agent can remotely start/stop chargers.
- Delivered fuzzy charger name resolution and real-time SSE streaming for operator dashboards, enabling new support staff to handle complex charger faults independently from day one.

Client AI Agent — Natural Language Analytics

- Designed a conversational analytics agent for 6,000+ EV station owners using LLM tool-calling over 15-20 parameterised PostgreSQL query functions, with owner ID auto-injected from session, making cross-account data access structurally impossible.
- Implemented Redis caching, nightly batch pre-aggregation, circuit breakers, per-owner rate limiting, and prompt regression tests blocking deployments below 95% accuracy, serving 1,700+ daily active users.
- Achieved near-zero time-to-insight for energy, session, and fault analytics, eliminating manual dashboard navigation for station owners.

Dynamic Tariff & ML Pricing System

- Built a nightly two-level ML pricing pipeline with network-wide demand forecasting (XGBoost, Random Forest, Facebook Prophet with auto best-model selection) and per-charger pricing via an activity classifier paired with a log-space energy regressor.

- Applied KMeans behavioural clustering on 36-feature charger profiles, p95 outlier clipping, log1p transformation, and asymmetric sigmoid surge mapping within regulatory caps (+12%/-6%), delivering approximately 10% revenue uplift.

Customer Retention & Churn Reduction

- Built a monthly churn detection system with energy-drop threshold segmentation, rolling-percentile lifecycle clustering across 5 tiers, and a proactive calling console for customer care teams.
- Delivered a Power BI management dashboard and a care-team console with call outcome logging, recovering churned revenue at a fraction of new acquisition cost.

Cloud ETL Pipeline & Power BI Dashboard

- Designed a nightly AWS Glue + S3 Parquet + Athena serverless pipeline extracting EV transactions from MongoDB and transforming with PySpark; date partitioning reduced cloud query costs by approximately 95%.
- Built idempotent partition overwrite logic, a 48-hour lookback buffer, and schema-anomaly quarantine; Power BI auto-refreshes daily, processing 1M+ transactions nightly with 30% faster data processing.

Maintenance & Uptime Monitoring System

- Built a nightly Python batch job computing gap-based uptime/downtime per charger from MongoDB logs, handling schema divergence between AC and DC chargers, contributing to approximately 20% reduction in operational downtime.

Data Science Intern — ChargeMOD, Trivandrum, Kerala **Mar 2025 – May 2025**

- Developed an EV Assistant Chatbot using OpenAI function calling, embedding-based retrieval, dynamic entity extraction, and real-time SSE streaming for route optimisation and live charger discovery; the project evolved into the production Customer Support AI Agent.

RPA Data Analyst — Software Incubator Pvt. Ltd., Kerala **Jun 2022 – May 2024**

- Automated document extraction and data processing workflows using JavaScript, Puppeteer, and Kofax; built browser automation pipelines for structured web scraping and data indexing into MySQL databases.

ACADEMIC & RESEARCH PROJECTS

M.Tech Thesis — Lip Reading using Deep Learning

- Designed an audio-free visual speech recognition system using 3D-CNN spatiotemporal feature extraction and BiLSTM sequence decoding trained with CTC loss on the GRID Corpus, achieving a Character Error Rate of 2.95% and Word Error Rate of 10.11%. Technologies: Python, TensorFlow, OpenCV.

Other Academic Projects

- Fruit Picker Robot (computer-vision object detection with robotic-arm sorting); Fashion MNIST ANN Classifier (89% accuracy); Image-to-Cartoon Generation pipeline (edge detection and colour quantisation); 3D Robotic Arm Web Application (Three.js browser-based simulation).

Publications

- “Machine Learning Models for Gamma-Hadron Separation,” CREEST International Conference — Gradient Boosting ensemble for gamma-ray vs. hadronic background classification, outperforming SVM, AdaBoost, Naive Bayes, and Decision Tree baselines.

EDUCATION

- M.Tech in Computer Science (Digital Image Computing) — Kerala University, Karyavattom (2023 – 2025)
- B.Tech in Computer Science and Engineering — University College of Engineering, Karyavattom (2015 – 2019)

CERTIFICATIONS & ACTIVITIES

- The Complete Python Bootcamp: From Zero to Hero — Udemy
- Data Visualization with Power BI — Great Learning
- Co-led an AI, Machine Learning & Robotics Bootcamp at Tella Academy for high school students